ModelIt!

Student Activity Pack

How LeBron Turns Food Into Dunks

The Cellular Science Behind Athletic Performance

Grade 8 | NGSS MS-LS1-5, MS-LS1-7 | Energy and Matter

STUDENT NAME _____ DATE _____

ModelIt! — Discovery Collective

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Table of Contents

ModelIt!

1 Pre-Assessment 3

2 Scoring Rubric 4

3 Component Analysis 5

4 Model Recording Page 6

5 Simulation Observations 7

6 Research & Extend 8

7 Reflection Questions 9

8 STEM Challenge 10

1 Pre-Assessment

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What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1 Where does the energy for your muscles come from?

2 What happens in your body when you eat food?

3 Why do you breathe harder when you exercise?

4 Draw the journey of a piece of pizza from your plate to energy in your muscles.

2 Scoring Rubric

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Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

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What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

Food Intake (Glucose) | Oxygen Supply | ATP Production | Athletic Output

EXTERNAL	INTERNAL

? Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

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Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? Explain ONE relationship in your model. What happens when one component changes?

5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: Resting State

Normal food intake, resting oxygen levels

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: Maximum Effort

High glucose, maximum oxygen delivery

MY PREDICTION

WHAT I OBSERVED

3 Scenario 3: No Fuel

Lock Food Intake to 0%

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

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Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

Lactic Acid — Builds up during intense exercise when oxygen is insufficient

Heart Rate — Increases to deliver more oxygen to working muscles

Hydration Level — Water is essential for chemical reactions in cells

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

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The Big Question

How does a chicken sandwich become a slam dunk? Use evidence from your simulation to support your answers.

1. **How is your body like a car engine that burns fuel?**

2. **Why do marathon runners 'hit the wall' and suddenly lose energy?**

3. **What's the connection between breathing and cellular respiration?**

4. **Why do athletes need to eat more than non-athletes?**

5. **How does your model explain why you feel tired after skipping lunch?**

8 STEM Challenge: Design an Athlete's Nutrition Plan

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YOUR ENGINEERING MISSION

Design an evidence-based nutrition and training plan for a student athlete to optimize their performance using cellular respiration science.

THE SCENARIO

Your school's basketball team is preparing for the championship. The coach hired your team to create a science-based nutrition plan for game day.

Engineering Design Thinking

1. WHAT FOODS WILL YOU INCLUDE IN THE PRE-GAME MEAL AND WHY?
2. HOW MANY HOURS BEFORE THE GAME SHOULD THE ATHLETE EAT?
3. WHAT SHOULD THE ATHLETE EAT DURING HALFTIME?
4. HOW WILL YOU MEASURE WHETHER YOUR PLAN IMPROVES PERFORMANCE?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! Explain your design using scientific evidence from the simulation:

Real Career Connection

Sports Nutritionists and Exercise Physiologists optimize athletic performance using science. They work with professional teams, Olympic athletes, and everyday people. They earn $55,000-$95,000/year.